

LINUX COMMANDS FOR DEVOPS

Essential Reference Guide & Practice Notes

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This document is a practical, hands-on reference of Linux commands practiced for DevOps workflows — covering file and directory operations, process management, permissions, networking, archiving, text processing, disk usage, system information, cron scheduling, and user management.

1. File Operations

No.	Command	Description
1	<code>ls</code>	Shows files and folders in the current directory.
2	<code>ls -R</code>	Lists files and folders recursively inside all subdirectories.
3	<code>ls -a</code>	Shows hidden files and directories.
4	<code>ls -al</code>	Displays file permissions, owner, group, size, date, and filename.
5	<code>cd directory</code>	Changes the current directory.
6	<code>cd ..</code>	Moves one directory up.
7	<code>pwd</code>	Displays the present working directory.
8	<code>cat > filename</code>	Creates a new file and allows entering content.
9	<code>cat file1.txt file2.txt > file3.txt</code>	Merges two files into one file.
10	<code>touch filename</code>	Creates an empty file.
11	<code>rm filename</code>	Deletes a file.
12	<code>cp app.py backup.py</code>	Copies a file.
13	<code>mv app.py project/</code>	Moves a file to another directory.
14	<code>mv old.txt new.txt</code>	Renames a file.
15	<code>find / -name app.py</code>	Searches for a file from the root directory.
16	<code>less /var/log/syslog</code>	Views a file page by page.
17	<code>head -5 app.log</code>	Shows the first 5 lines of a file.
18	<code>tail -f app.log</code>	Continuously displays new lines added to a file.

2. Directory Operations

No.	Command	Description
1	<code>mkdir project</code>	Creates a directory.
2	<code>mkdir -p project/src/java</code>	Creates parent and child directories in one command.
3	<code>tree project</code>	Displays the directory structure in a tree format.

No.	Command	Description
4	<code>rmdir project</code>	Deletes an empty directory.
5	<code>rm -rf project</code>	Deletes a non-empty directory (-r recursive, -f force).
6	<code>cp -r source destination</code>	Copies a directory and all its contents.
7	<code>find / -type d -name logs</code>	Finds a directory by name.
8	<code>find / -type f -name app.py</code>	Finds a file by name.

3. Process Operations

No.	Command	Description
1	<code>ps</code>	Shows currently running processes.
2	<code>ps -ef</code>	Displays detailed information about all running processes.
3	<code>top</code>	Displays real-time CPU and memory usage.
4	<code>kill PID</code>	Terminates a process by PID.
5	<code>kill -9 PID</code>	Forcefully terminates a process.
6	<code>pkill process_name</code>	Kills a process by its name.
7	<code>bg</code>	Resumes a suspended job in the background.
8	<code>fg</code>	Brings the latest background job to the foreground.
9	<code>fg %2</code>	Brings a specific background job to the foreground.
10	<code>renice</code>	Changes the priority of a running process.
11	<code>command &> output.log</code>	Redirects both output and errors to a file.
12	<code>command 1> output.log</code>	Redirects only standard output.
13	<code>command 2> error.log</code>	Redirects only error output.

4. File Permissions

No.	Command	Description
1	<code>ls -l app.sh</code>	Displays file permissions and ownership.
2	<code>chmod 755 filename</code>	Changes file permissions using octal notation.
3	<code>chown user filename</code>	Changes the file owner.
4	<code>sudo chown user:group filename</code>	Changes the file owner and group.

No.	Command	Description
5	<code>sudo chgrp developers app.log</code>	Changes the group ownership.
6	<code>chmod +x deploy.sh</code>	Makes a file executable.
7	<code>sudo chown -R ubuntu:ubuntu /opt/myapp</code>	Recursively changes owner and group.

5. Networking Commands

No.	Command	Description
1	<code>mtr google.com</code>	Combines ping and traceroute for network diagnostics.
2	<code>nmap 192.168.1.10</code>	Scans hosts, ports, and services.
3	<code>ping google.com</code>	Checks network connectivity.
4	<code>whois google.com</code>	Displays domain registration information.
5	<code>dig google.com</code>	Queries DNS records.
6	<code>netstat -pnltn</code>	Shows active connections and listening ports.
7	<code>ifconfig / ip addr</code>	Displays network interface information.
8	<code>ssh ubuntu@192.168.1.10</code>	Connects securely to a remote server.
9	<code>scp app.jar ubuntu@192.168.1.10:/home/ubuntu/</code>	Copies files securely over SSH.
10	<code>wget URL</code>	Downloads files from the internet.
11	<code>curl URL</code>	Sends HTTP requests and displays responses.
12	<code>traceroute google.com</code>	Shows the network path to a destination.
13	<code>ss -tuln</code>	Displays socket and listening port information.

6. File Management (Archiving & Compression)

No.	Command	Description
1	<code>mkdir project</code>	Creates a project directory.
2	<code>touch project/app.py</code>	Creates files.
3	<code>tar -cvf project.tar project/</code>	Creates a TAR archive.
4	<code>tar -xvf project.tar</code>	Extracts a TAR archive.

No.	Command	Description
5	<code>gzip project.tar</code>	Compresses a file using Gzip.
6	<code>gzip -d project.tar.gz</code>	Decompresses a Gzip file.
7	<code>zip -r project.zip project/</code>	Creates a ZIP archive.
8	<code>unzip project.zip</code>	Extracts a ZIP archive.
9	<code>tar -jcvf project.tar.bz2 project/</code>	Creates a Bzip2 archive.
10	<code>tar -jxvf project.tar.bz2</code>	Extracts a Bzip2 archive.
11	<code>tar -czvf website-backup.tar.gz /var/www/html</code>	Creates a compressed website backup.
12	<code>tar -xzvf website-backup.tar.gz</code>	Extracts a .tar.gz archive.

7. Text Processing

No.	Command	Description
1	<code>grep "ERROR" app.log</code>	Searches text in a file.
2	<code>grep -n "ERROR" app.log</code>	Shows matching line numbers.
3	<code>grep -c "ERROR" app.log</code>	Counts matching lines.
4	<code>grep -r "database" project/</code>	Searches recursively in directories.
5	<code>ps -ef grep java</code>	Filters command output.
6	<code>echo "Hello"</code>	Displays text or variable values.
7	<code>sed 's/old/new/g' file</code>	Replaces text in output.
8	<code>sed -i 's/old/new/g' file</code>	Permanently replaces text in a file.
9	<code>diff old.conf new.conf</code>	Compares two files.
10	<code>wc notes.txt</code>	Counts lines, words, and characters.
11	<code>wc -l file</code>	Counts lines only.
12	<code>wc -w file</code>	Counts words only.
13	<code>wc -c file</code>	Counts characters/bytes.
14	<code>awk '{print \$1}' users.txt</code>	Prints the first column.
15	<code>cut</code>	Extracts specific columns or characters.
16	<code>tail -20 access.log</code>	Shows the last 20 lines.

No.	Command	Description
17	<code>head -20 file.txt</code>	Shows the first 20 lines.

8. Disk Usage Commands

No.	Command	Description
1	<code>df</code>	Shows disk usage of mounted file systems.
2	<code>df -h</code>	Shows disk usage in human-readable format.
3	<code>du</code>	Shows the size of files and directories.
4	<code>du -sh project</code>	Shows the total size of the project directory.
5	<code>du -sh *</code>	Shows the size of each file and directory.
6	<code>free</code>	Displays RAM and swap usage.
7	<code>free -h</code>	Displays memory usage in human-readable format.
8	<code>whereis git</code>	Shows the binary, source, and manual page locations.

9. System Information

No.	Command	Description
1	<code>date</code>	Displays the current date and time.
2	<code>date "+%d-%m-%Y %H:%M:%S"</code>	Displays formatted date and time.
3	<code>cal</code>	Shows the current month's calendar.
4	<code>cal 2026</code>	Shows the calendar for the entire year.
5	<code>uptime</code>	Displays system uptime and load average.
6	<code>w</code>	Shows logged-in users and their activities.
7	<code>who</code>	Lists logged-in users.
8	<code>whoami</code>	Displays the current username.
9	<code>uname -a</code>	Displays complete system information.
10	<code>uname -r</code>	Shows the kernel version.
11	<code>uname -m</code>	Displays the system architecture.

10. Job Scheduling (Cron)

No.	Command	Description
1	<code>crontab -l</code>	Lists all cron jobs.
2	<code>crontab -e</code>	Opens the cron file for editing.
3	<code>crontab -r</code>	Deletes all cron jobs.
4	<code>crontab -v</code>	Shows the last edit time (supported on some systems).
5	<code>crontab filename</code>	Loads cron jobs from a file.
6	<code>@reboot</code>	Runs once after system startup.
7	<code>@hourly</code>	Runs every hour.
8	<code>@daily</code>	Runs once every day.
9	<code>@weekly</code>	Runs once every week.
10	<code>@monthly</code>	Runs once every month.
11	<code>@yearly</code>	Runs once every year.

11. User Management Commands

No.	Command	Description
1	<code>whoami</code>	Displays the current logged-in username.
2	<code>who</code>	Shows all currently logged-in users.
3	<code>w</code>	Shows logged-in users and what they are doing.
4	<code>id username</code>	Displays the user's UID, GID, and group information.
5	<code>users</code>	Lists the usernames of currently logged-in users.
6	<code>groups username</code>	Shows the groups a user belongs to.
7	<code>sudo adduser john</code>	Creates a new user with a home directory and password prompt.
8	<code>sudo useradd john</code>	Creates a new user (basic account).
9	<code>sudo passwd john</code>	Sets or changes a user's password.
10	<code>sudo usermod -aG sudo john</code>	Adds a user to the sudo group.
11	<code>sudo usermod -aG developers john</code>	Adds a user to the developers group.
12	<code>sudo groupadd developers</code>	Creates a new group.
13	<code>sudo groupdel developers</code>	Deletes a group.
14	<code>sudo userdel john</code>	Deletes a user account.

No.	Command	Description
15	<code>sudo userdel -r john</code>	Deletes a user and their home directory.
16	<code>sudo passwd -l john</code>	Locks a user account.
17	<code>sudo passwd -u john</code>	Unlocks a user account.
18	<code>sudo chage -l john</code>	Displays password aging information.
19	<code>sudo chage -E 2026-12-31 john</code>	Sets an account expiration date.
20	<code>sudo su -</code>	Switches to the root user.
21	<code>su - john</code>	Switches to another user account.
22	<code>exit</code>	Returns to the previous user or shell.
23	<code>sudo visudo</code>	Safely edits the sudoers file.
24	<code>cat /etc/passwd</code>	Displays all user accounts.
25	<code>cat /etc/group</code>	Displays all groups.
26	<code>cat /etc/shadow</code>	Displays encrypted password information (root only).